

CargoFlow Sentinel

From shipment alert to business-impact action | 3-agent system built on Microsoft Foundry | 2026 Microsoft Virtual Agent-a-thon
Code & full evidence: github.com/ssfathimaindia-wq/cargoflow-sentinel

The challenge: Modern logistics platforms can already detect delays, ETA changes and cold-chain breaches. The harder question is: what does this exception mean to the business, and what should happen next?

REAL-WORLD EXAMPLE: SHP-1003

ROUTE Gdansk → Berlin	CARRIER BalticRoute GmbH	CUSTOMS DELAY +12.8 hrs
COLD-CHAIN 9.4°C vs 6.0°C limit	LINKED PO PO-88203	BUSINESS CONTEXT €41,200 · production-dependent

HOW THIS IS HANDLED TODAY VS. CARGOFLOW SENTINEL

Today's connected logistics landscape 1. Shipment / TMS / carrier feed — GPS, carrier APIs and milestone events expose location, ETA and customs status. 2. Cold-chain telemetry — IoT / reefer sensors report temperature. A rule can flag 9.4°C against a 6.0°C threshold. 3. Exception alert — A control tower or visibility platform can surface: 12.8-hour delay + temperature excursion. 4. Business context — ERP/WMS/order systems hold PO, customer, value, inventory and production dependencies. Depending on system maturity, this context may be integrated automatically or require cross-system investigation. ----- <i>Result: systems are strong at producing operational signals; the response still depends on how well exception, business context and action workflows are connected.</i>	CargoFlow Sentinel agent approach Agent 1 — Exception Detection — consumes structured shipment facts and identifies SLA / cold-chain exceptions. Agent 2 — Root-Cause & Action — interprets the combined signals, diagnoses the likely cause and recommends an actionable response. Conditional gate — only CRITICAL cases continue. Lower-severity cases are logged without unnecessary ERP escalation. Agent 3 — ERP Reconciliation — adds business context: linked PO, customer, order value and production dependency. ----- <i>Result: "temperature exceeded" becomes: critical shipment exception affecting a €41,200 production-dependent order — urgent escalation.</i>
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ARCHITECTURE & DECISION FLOW

Real-World Signals

Carrier / GPS · Customs events · IoT temperature

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Agent 1

Exception Detection — SLA + cold-chain

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Agent 2

Root Cause & Action — diagnose + recommend

→

Critical?

No → log · Yes → ERP

→

Agent 3

ERP Reconciliation — PO · customer · value · impact

WHY AGENTS?

Hard thresholds such as 9.4°C > 6.0°C stay deterministic. Agents add value in interpretation, cross-system reasoning, prioritisation and recommended action.

RELIABILITY & PRIVACY

Every agent call retries with exponential backoff, but only on transient errors (rate limits, timeouts, 5xx) — never on auth/bad-request errors. Data today is synthetic with no PII; a live D365FO connection would add GDPR-appropriate handling before go-live.

PROTOTYPE SCOPE & PROOF

Three deployed Foundry agents; conditional branch verified on two independent paths (Python SDK + Foundry workflow). 4.5/5 hand-written eval; 86% coherence / 100% fluency on a second, Foundry-native synthetic eval.

Business value & fit: the value is in latency, not automation — the same fault costs nothing caught early, a write-off caught late. Best fit: shipment volume past manual monitoring, cold-chain or JIT-production cargo, and an ERP that already links purchase order to customer — Microsoft Dynamics 365 Finance & Operations is the natural target.

Production vision: replace representative mock feeds with live carrier/IoT tracking and D365FO/ERP connections. The architecture stays the same; the operational systems become the source of truth.

MAPS TO THE JUDGING LENS		
INNOVATION The conditional gate itself — business-impact-aware, not another linear alert pipeline.	USABILITY Judged output (4.5/5) plus two independently verified execution paths — SDK and Foundry workflow.	IMPACT Named cost categories tied to real ERP fields, not abstract efficiency claims.